MIL-I-19632B(SHIPS)
14 April 1960
SUPERSEDING
MIL-I-19632A(SHIPS)
25 September 1959

MILITARY SPECIFICATION

INSULATION, ELECTRICAL, DIELECTRIC BARRIER, LAMINATED, PLASTIC FILM AND RAG PAPER

1. SCOPE

1.1 Scope. - This specification covers insulation used primarily as dielectric barriers in class A electrical machinery.

1.2 Classification. -

1.2.1 Compositions. - The electrical insulation shall be of the following compositions, as specified (see 6.2):

Composition FR - One layer of polyethylene terephthalate film or sheet laminated to one layer of 100 percent rag paper (duplex).
Composition RFR - One layer of 100 percent rag paper laminated to each side of a polyethylene terephthalate film or sheet (triplex).
Composition RoFRo - One layer of rope paper laminated to each side of a polyethylene terephthalate film or sheet (triplex).

1.2.2 Types. - The electrical insulation shall be of the following types as specified (see 6.2).

Types:

FR-7
FR-9
FR-10
FR-12a
FR-12b
FR-15
RFR-13
RFR-16
RFR-19
RFR-22a
RFR-22b
RoFRo-4

1.2.3 Forms. - The electrical insulation shall be of the following forms, as specified (see 6.2):

Form R - Rolls, trimmed, 35 to 35-1/2 inches in width, in lengths of 50 yards and 200 yards.

Form S - Sheets, 36 inches wide by 36 inches in length in packages of 50 and 250 sheets.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
UU-P-268 - Paper, Kraft, Wrapping.
UU-T-111 - Tape; Paper, Gummed (Sealing and Securing).
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.
PPP-B-636 - Boxes, Fiber.
PPP-T-76 - Tape, Pressure-Sensitive, Adhesive, Paper, Water Resistant.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-I-695 - Insulation, Electrical, Paper (Slot-Cell).
MIL-B-10377 - Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL-L-10547 - Liners, Case, Waterproof.

STANDARDS

FEDERAL
FED-STD-601 - Rubber: Sampling and Testing.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 5970

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MILITARY (cont'd.)
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Material. -

3.1.1 Plastic film or sheet. - The polyethylene terephthalate film or sheet shall conform to type G of Specification MIL-I-631.

3.1.2 Rag paper. - The rag paper shall be 100 percent rag paper conforming to type R of Specification MIL-I-695.

3.1.3 Rope paper. - The rope paper shall be regular electrical grade rope paper.

3.1.4 Composite insulation. - The composite insulation shall be uniform in texture, density, and finish, without breaks, cracks, wrinkles, or flaws of any kind. The paper shall be well formed and there shall be no tendency towards splitting or separation of the laminations. The layers shall be cemented together by a flexible insulating binder. The insulation shall be free from electrically conductive particles, dirt, and other foreign material that may affect the serviceability of the finished product.

3.2 Construction and thickness. - The electrical insulation shall conform to the construction and thickness shown in table I (see 6.2).

Table I - Construction and thickness.

Type	Nominal combined insulation thickness (see 6.2)	Tolerance of average combined thickness from nominal thickness		Nominal thickness (inch) and component (designation)[1]		
		Minus	Plus	First layer	Second layer	Third layer
	Inch	Inch	Inch			
FR-7	0.007	0.0005	0.001	[2] 0.002F	0.005R	------
FR-9	.009	.0005	.001	[2] .002F	.007R	------
FR-10	.010	.0005	.001	[2] .005F	.005R	------
FR-12a	.012	.0005	.001	[2] .002F	.010R	------
FR-12b	.012	.0005	.001	.005F	.007R	------
FR-15	.015	.0005	.001	.005F	.010R	------
RFR-13	.013	.0005	.002	.005R	[2] .003F	0.005R
RFR-16	.016	.0005	.002	.007R	[2] .002F	.007R
RFR-19	.019	.0005	.0025	.007R	[2] .002F	.010R
RFR-22a	.022	.0005	.0025	.007R	[2] .005F	.010R
RFR-22b	.022	.0005	.0025	.010R	[2] .002F	.010R
RoFRo-4	.004	.0000	.001	.001Ro	[2] .002F	.001Ro

[1] Designations:
F - Plastic film or sheet.
R - Rag paper.
Ro - Rope paper.

[2] The tolerance on the plastic film or sheet component thickness shall be plus or minus 10 percent of the nominal thickness.

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3.3 Physical requirements. - The electrical insulation shall conform to the physical requirements shown in table II.

Table II - Physical properties.

Property	Condition	Unit of value	Value required											
			Type FR-7	Type FR-9	Type FR-10	Type FR-12a	Type FR-12b	Type FR-15	Type RFR-13	Type RFR-16	Type RFR-19	Type RFR-22a	Type RFR-22b	Type RoFRo-4
Dielectric strength	C-96/23/50 Straight	Volts per mil (minimum average)	1250	1000	1200	750	1000	800	900	650	625	800	600	2,000
	C-96/23/96 Straight	Volts per mil (minimum average)	880	700	1080	530	900	720	750	500	425	650	350	1,300
	C-96/23/50 Bent lengthwise	Volts per mil (minimum average)	1250	1000	1080	700	900	640	900	650	625	700	600	2,000
	C-96/23/50 Bent crosswise	Volts per mil (minimum average)	1250	1000	1080	700	900	640	900	650	625	700	600	2,000
	E-168/130 Bent lengthwise	Volts per mil (minimum average)	1000	800	860	550	720	510	700	580	560	600	540	2,000
Volume resistivity	C-96/23/96 ----------	Ohm-cms. (minimum average)	10^{14}	10^{14}	10^{14}	10^{14}	10^{14}	10^{14}	10^{11}	10^{11}	10^{11}	10^{11}	10^{11}	10^{13}
Tensile strength	C-96/23/50 Lengthwise	Pounds per inch width (minimum average)	90	115	130	160	160	190	180	220	300	325	350	50
	C-96/23/50 Crosswise	Pounds per inch width (minimum average)	55	75	90	110	110	140	100	100	120	170	140	50
Tear resistance[1] (slot cell)	C-96/23/50 Lengthwise	Pounds (minimum average)	15	17	30	20	35	40	30	25	30	50	35	11
	C-96/23/50 Crosswise	Pounds (minimum average)	15	17	30	20	35	40	20	20	25	40	25	14

[1] The polyethylene terephthalate component shall not tear when tested uncuffed.

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3.4 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. -

4.2.1 Lot. - All material of the same type and form manufactured under essentially the same conditions within a period of 15 days and offered for delivery at the same time shall be considered a lot for purposes of acceptance inspection.

4.2.2 Sampling for visual and dimensional examination. - A random sample shall be selected from each lot of material offered for examination of visual and dimensional characteristics in accordance with Standard MIL-STD-105 at inspection level I. The Acceptable Quality Level shall be 1.5 percent defective. Both the lot size and sample size shall be stated in terms of linear yards; however, the sample number of yards of material shall be distributed in such a way that approximately five yards shall be examined from any one roll or package.

4.2.3 Sampling for acceptance inspection. - One fifth of the rolls or packages selected for the visual and dimensional examination shall be subjected to the acceptance inspection specified in table III. The smallest number of rolls or packages selected for any sample shall be two.

Table III - Acceptance inspection.

Test	Requirement paragraph	Test reference	Minimum number of determinations per sample
Thickness, Condition C-96/23/50	3.2	4.6.1	10
Dielectric strength:			
Straight, Condition C-96/23/50	3.3	4.6.2	10
Straight, Condition C-96/23/96	3.3	4.6.2	10
Bent lengthwise, Condition C-96/23/50	3.3	4.6.2	10
Bent crosswise, Condition C-96/23/50	3.3	4.6.2	10
Bent lengthwise, Condition E-168/130	3.3	4.6.2	10
Volume resistivity, Condition C-96/23/96	3.3	4.6.3	3
Tensile strength:			
Lengthwise, Condition C-96/23/50	3.3	4.6.4	5
Crosswise, Condition C-96/23/50	3.3	4.6.4	5
Tear resistance (slot cell):			
Lengthwise, Condition C-96/23/50	3.3	4.6.5	5
Crosswise, Condition C-96/23/50	3.3	4.6.5	5

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4.3 Acceptance inspection. -

4.3.1 Visual and dimensional examination. - Each of the sample yards of material selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with the requirements of this specification not involving tests. Any sample yard containing one or more visual or dimensional defects shall be considered defective, and if the number of defective yards exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. In the case of rolls, the outer layer of the rolls shall be discarded and the next five linear yards shall be examined.

4.3.2 Acceptance tests. - Acceptance tests shall be conducted on samples of material selected in accordance with 4.2.3. Two linear yards from each sample roll or package shall be subjected to the tests specified in table III. Results shall be averaged for each sample roll or package represented to determine conformance with this specification. If the average for any sample roll or package fails to conform with any requirement, the lot shall be rejected.

4.4 Test conditions. -

4.4.1 Conditions. - Unless otherwise specified herein, all tests shall be conducted under conditions of 23° ± 1.1° centigrade (C.) and 50 ± 2 percent relative humidity (r.h.).

4.4.2 Designation. - Conditioning procedures shall be designated as follows:

(a) A capital letter indicating the general condition of the specimen: "C" indicates humidity and "E" indicates temperature conditioning.
(b) A number indicating in hours the duration of the conditioning.
(c) A number indicating in degrees C. the conditioning temperature.
(d) A number indicating relative humidity whenever relative humidity is controlled.

The numbers shall be separated from each other by slant marks, and from the capital letter by a dash.

Examples:
C-96/23/50 - 96 hours, 23° C., 50 percent relative humidity.
C-96/23/96 - 96 hours, 23° C., 96 percent relative humidity.
E-168/130 - 168 hours, 130° C.

4.5 Visual and dimensional examination. - The electrical insulation shall be examined to determine conformance with the requirements of 3.1, 3.2, and 3.4

4.6. Test procedures. -

4.6.1 Thickness. - The thickness shall be determined by the use of a dial type micrometer. The total thickness of the material shall be measured. The composite shall be soaked in water and the paper layer or layers peeled off. The adhesive adhering to the plastic film or sheet shall then be removed by soaking the plastic film or sheet in xylol and rubbing with cloth. The thickness of the plastic film or sheet shall then be measured. The total thickness of the paper layer or layers shall be determined by the difference. The lower or fixed anvil of the dial mircometer shall consist of a flat steel plate 2 inches in diameter; the upper or movable member shall consist of a foot having a flat steel face measuring 1/4 inch in diameter, and shall be arranged so as to move with its face exactly parallel to the face of the anvil at all times. The movable member shall be of such weight, and shall be attached to the operating rod of the micrometer in such a manner, that 20 ounces dead weight shall be applied to all thicknesses of samples being measured. The thickness shall be determined as the average of at least 10 readings taken at representative points on the material. The maximum, minimum, and average values shall be reported.

4.6.2 Dielectric strength. -

4.6.2.1 Test equipment. - The test equipment shall consist of a high voltage transformer rated at not less than 2 kilovolt-amperes and the necessary auxiliary equipment for applying, controlling, and measuring the test voltage. The power supply shall consist of an alternating-current source having as nearly a true sine wave as possible at a frequency not exceeding 100 cycles per second.

4.6.2.2 Voltage control. - The high-tension voltage taken from the secondary of the testing transformer shall be capable of being raised gradually from any point and in no case more than 500 volts at a step. The control may be made by generator field regulation with an induction regulator, or with a variable ratio autotransformer. Any method of controlling the voltage which does not distort the wave more than 10 percent from a sinusoidal shape is satisfactory.

4.6.2.3 Voltage measurement. - The voltage may be measured by any method which gives root-mean-square values, preferably by means of a voltmeter connected to a tertiary coil in the testing transformer, or to a separate step-down instrument potential transformer. A voltmeter on the low tension side of the transformer is satisfactory if the ratio of the transformation does not change under any test condition. An electrostatic voltmeter properly calibrated in the high tension circuit is also satisfactory. A spark gap may be used to check the readings at very high potentials.

4.6.2.4 Medium and rate of voltage rise. - The tests shall be conducted in air by the short-time method. Starting at zero, the voltage shall be increased uniformly to a breakdown, at a rate of 500 volts per second.

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4.6.2.5 Procedure - straight specimens. - A sufficient number of specimens 4 inches by 4 inches shall be cut to permit making 10 breakdowns after each of the following conditions:

C-96/23/50 (on straight specimens).
C-96/23/96 (on straight specimens).

The test electrodes shall consist of metal cylinders 2 inches in diameter and not less than 1/2 inch in length with edges rounded to a radius of 1/4 inch. The total weight on the sample shall be approximately 1 pound.

4.6.2.6 Prodedure - bent specimens. - A sufficient number of specimens 3 by 3-1/2 inches shall be cut to permit making 10 breakdowns after each of the following conditions:

C-96/23/50 (on specimens bent lengthwise).
C-96/23/50 (on specimens bent crosswise).
E-168/130 (on specimens bent lengthwise).

The specimens shall be bent 90 degrees in the V-shaped jig shown on figure 1 with a paper layer in tension. (In the case of types RFR-19 and RFR-22a, the thicker paper layer shall be in tension. The same side of the composite shall be in tension for the various tests.) Breakdowns shall be made using 90-degree, V-shaped electrodes designed to hold the specimens in a bent position. These electrodes are shown on figure 2. All edges of the bending jig and electrodes are rounded to approximately 1/64 inch radius. Oven aged specimens shall be clamped on a rectangular aluminum channel to hold them in a bent position during aging. After aging, the channels shall be removed from the oven and allowed to return to room condition. The specimens shall then be removed from the channel and short-time breakdowns made using the V-shaped electrodes.

4.6.2.7 Calculation of dielectric strength. - The dielectric strength shall be computed by dividing the average dielectric breakdown by the average thickness, and shall be reported in volts per mil. The average thickness shall be determined as specified in 4.6.1.

4.6.3 Volume resistivity. -

4.6.3.1 Apparatus. - The apparatus shall consist of a source of direct-current potential of 100 to 200 volts, a galvanometer with suitable shunts having a minimum sensitivity of 0.0005 microampere per millimeter at 1 meter-scale distance, a calibrated resistance, reversing switches, and keys. Any other apparatus may be used provided the same accuracy is obtained and the same voltage is applied.

4.6.3.2 Test specimens. - The size of the test specimens shall be 4 by 4 inches. Three specimens shall be tested after Condition C-96/23/96.

4.6.3.3 Test electrodes. - One electrode shall be of metal foil, 3 inches in diameter, on one surface of the specimen. The electrode on the other surface shall be of metal foil 2-5/8 inches in diameter. A concentric, metal-foil, guard ring electrode shall be used and shall be 1/8 inch wide and 3 inches in outside diameter and be spaced 1/16 inch from the inner electrode. The electrodes shall be applied to the specimen by means of a very thin film of petrolatum and shall be thoroughly rolled to exclude air. The electrodes shall be accurately centered with respect to each other.

4.6.3.4 Procedure. - The volume resistance between the guarded and unguarded electrodes shall be measured after voltage has been applied for 1 minute.

4.6.3.5 Calculation. - Volume resistivity, r, in ohm-centimeters, shall be calculated as follows:

$$r = \frac{RA \times 10^6}{t}$$

Where:
R = volume resistance in megohms.
A = area of guarded electrode in square centimeters.
t = average thickness of specimen in centimeters.

The average volume resistivity shall be reported.

4.6.4 Tensile strength. -

4.6.4.1 Apparatus. - Tensile strength shall be determined on an approved pendulum-type or constant rate of jaw travel testing machine. The machine shall be power-driven.

4.6.4.2 Specimens. - The test specimens shall be cut using die I of method 4111 of Standard FED-STD-601. The surface of the specimen shall be smooth and free from scratches. Five specimens cut lengthwise and five cut crosswise of the sheet shall be used after Condition C-96/23/50.

4.6.4.3 Procedure. - The test specimens shall be placed in the testing machine and stretched until they break. The rate of separation of the jaws of the machine shall be 12 plus or minus 1/2 inches per minute. The tensile strength shall be reported in pounds per inch width.

4.6.5 Tear resistance (slot-cell). -

4.6.5.1 Apparatus. - The slot-cell tear resistance shall be determined using the tensile strength apparatus specified in 4.6.4.1 with specimens mounted in the slot-cell tear resistance jig shown on figure 3.[1]

[1] Working drawings of this jig are available from Material Laboratory (Code 955), New York Naval Shipyard, Brooklyn 1, N.Y.

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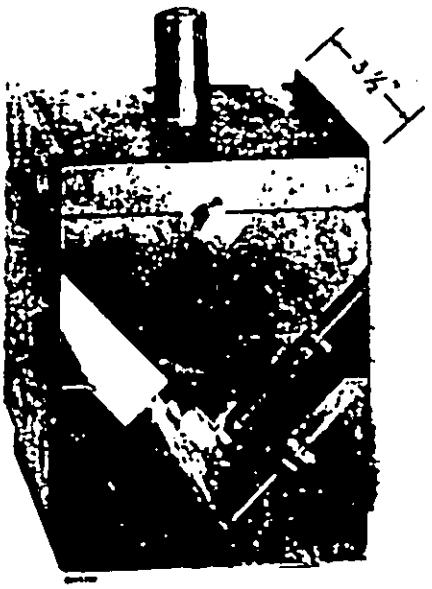


FIGURE 1 - BENDING JIG 90° V-SHAPED



FIGURE 2 - DIELECTRIC TEST ELECTRODES 90° V-SHAPED

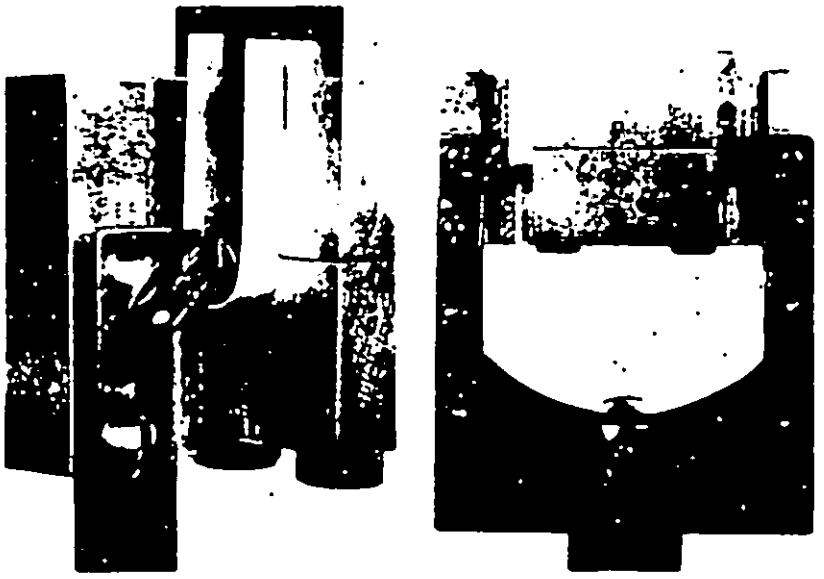


FIGURE 3 - SLOT-CELL TEAR RESISTANCE JIG SPECIMEN IN POSITION TO START TEAR

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4.6.5.2 Specimens. - Five specimens shall be torn in the tearing jig in the lengthwise direction and five specimens shall be torn in the tearing jig in the crosswise direction after Condition C-96/23/50. The directions shall be reported as the directions in which the tears run. The specimens shall be 2-1/2 by 3 inches. The specimens shall be inserted and aligned in the tearing jig with a paper layer in tension. (In the case of types RFR-19 and RFR-22a, the thicker paper layer shall be in tension. The same side of the composite shall be in tension for the various tests.) The specimen holder shall be clamped to the upper jaw of the tensile machine. The tearing plate together with the attached guide plate shall then be clamped to the lower jaw. The tearing edges which come in contact with the specimen have a radius of curvature of 1/32 inch.

4.6.5.3 Procedure. - The maximum force in pounds developed in tearing through the edges of the specimen extending beyond the holder at a constant rate of 12 inches per minute shall be recorded on a chart. The average tear resistance in pounds for each direction of tear shall be calculated and reported.

4.7 Inspection of preparation for delivery. - Sample packages shall be selected and inspected in accordance with Specification MIL-P-116 to verify conformance with Section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. - Insulation furnished in rolls or sheets as specified (see 6.2), shall be packaged in accordance with method III of Specification MIL-P-116.

5.1.1.1 Rolls. - Rolls shall be individually wrapped in minimum 25-pound basis-weight kraft paper conforming to Specification UU-P-268 and securely taped with gummed kraft tape conforming to Specification UU-T-111. Each wrapped roll shall be completely circled with the kraft tape at a minimum of four places equidistant along its length, with the tape overlapping itself at least 2 inches.

5.1.1.2 Sheets. - Sheets shall be interleaved with paper to prevent abrasion. The sheets shall be packaged in a fiber box conforming to Specification PPP-B-636, or shall be wrapped in minimum 25-pound basis-weight kraft paper conforming to Specification UU-P-268 and securely taped with tape conforming to Specification UU-T-111.

5.1.2 Level C. - Insulation shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. - Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. - Insulation, packaged as specified (see 6.2), shall be packed in overseas type, wood-cleated fiberboard, nailed wood, fiber, wirebound wood, wood-cleated veneer paper overlaid, or wood-cleated plywood boxes conforming to Specification PPP-B-591, PPP-B-621 (class 2), PPP-B-636 (class 2), PPP-B-585 (class 3 use), MIL-B-10377 or PPP-B-601, respectively, at the option of the contractor. Shipping containers shall have caseliners conforming to Specification MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to Specification MIL-L-10547. Caseliners for boxes conforming to Specification PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with minimum 1-1/2 inch wide tape conforming to Specification PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiber boxes may be banded with tape in accordance with the Appendix to Specification PPP-B-636 gross weight of wood boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 Level B. - Insulation, packaged as specified (see 6.2), shall be packed in domestic type wood-cleated fiberboard, nailed wood, wirebound wood, cleated plywood, or wood-cleated veneer paper overlaid boxes or class 2 fiber boxes conforming to Specification PPP-B-591, PPP-B-621 (class 1), PPP-B-585 (class 1 or 2 use), PPP-B-601, MIL-B-601, MIL-B-1037 or PPP-B-636, respectively, at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 Level C. - Insulation, packaged as specified (see 6.2), shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.3 Marking. - In addition to any special marking required by the contract or order or herein, interior package and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. - The materials covered by this specification are class A electrical insulations with continuous operating temperature limit of 105° C. They are intended for use as slot-cell insulations in electrical machinery where high tear resistance and excellent resistance to high humidity are required.

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6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Composition required (see 1.2.1).
(c) Type required (see 1.2.2).
(d) Form required (see 1.2.3).
(e) Length of roll in yards, if to be furnished in form R (see 1.2.3).
(f) Number of sheets in package, if to be furnished in form S (see 1.2.3).
(g) Nominal thickness of combined insulation and each component (see 3.2).
(h) Level of preservation and packaging level of packing required (see 5.1 and 5.2).

6.3 Approximate areas per pound. - The approximate area per pound of the materials covered by this specification are shown in table IV.

Table IV - Areas per pound (approximate).

Type	Nominal combined insulation thickness	Area in square yards per pound
	Inch	
FR-7	0.007	2.15
FR-9	.009	1.69
FR-10	.010	1.50
FR-12a	.012	1.28
FR-12b	.012	1.31
FR-15	.015	1.10
RFR-13	.013	1.16
RFR-16	.016	0.952
RFR-19	.019	.80
RFR-22a	.022	.71
RFR-22b	.022	.697
RoFRo-4	.004	4.09

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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